# Gradients with a numerical check

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

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## Prepare the purpose, prerequisites and materials

> Prepare Gradients with a numerical check: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Estimate one **partial derivative** while other parameters stay fixed

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate an average rate of change over a stated interval

Reference example, not the only acceptable wording: Teacher explanation: Average slope is (0 - 4) / 2 = -2. The local slope at 2 is zero. They describe different locations or intervals.

### Parameters and architecture

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish architectural choices from parameter values

Reference example, not the only acceptable wording: Teacher explanation: There are 8 weights and 2 biases, total 10.

### Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Look for: Trace identity ReLU and sigmoid on a numerical score

Reference example, not the only acceptable wording: Teacher explanation: Identity gives -2, 0, 2; ReLU gives 0, 0, 2. ReLU retains the positive score itself; it is not a step classifier.

Use two 45-minute sessions to separate coordinate probes from combined changes. The first **gradient** check is deliberately an affine model with squared loss, avoiding hidden-layer calculus.

Introduce the chain-rule reasoning in words before showing the compact formula.

### Outcomes

Estimate one **partial derivative** while other parameters stay fixed

Assemble a **gradient** in a declared parameter order

Compare numerical and formula based **gradient**s with a **tolerance**

### Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

### Prerequisite answer

The two evaluation points lie one step on either side of the centre, so their separation is two steps.

### End goal for the pictorial case

By the end, I can explain **gradient**s with a numerical check using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

# Explain and model the mechanism

> Model Gradients with a numerical check using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Use one example with **input** x = 2 and target t = 3. The model predicts y = wx + b, and loss L(w, b) = (2w + b - 3) squared. At w = 1, b = 0, **prediction** is 2, residual is -1 and loss is 1.

There are two parameters in order [w, b].

To estimate the partial slope for w, change w only and keep b = 0. To estimate the partial slope for b, change b only and keep w = 1.

Moving both at once measures change along a combined direction; it does not isolate either **partial derivative**.

The **gradient** is [-4, -2] in order [w, b]. Both entries refer to the same original parameter vector, not to a partially updated model.

## Explain Gradients with a numerical check · 1

The helper assumes finite numerical **input**s and a positive nonzero step. The formula follows the squared-loss rule: local loss change is twice the residual times the **prediction** sensitivity.

Changing w by a small amount changes **prediction** twice as much here; changing b changes **prediction** by the same amount.

Exact floating-point equality is not required. Record the step and a **tolerance**, then try more than one parameter setting.

A passing numerical check supports agreement for those tested points; it does not prove all possible **input**s.

| Term | Meaning |
| --- | --- |
| Partial derivative | A local slope with respect to one parameter while others are held fixed |
| Gradient | The ordered vector of partial derivatives of a scalar function |
| Finite difference | An estimate of change computed from nearby function values |

| Term | Meaning |
| --- | --- |
| Tolerance | A stated allowance for numerical disagreement |

### Two separate probes at one parameter vector

| PARAMETER BEING PROBED | HELD FIXED | GRADIENT ENTRY |
| --- | --- | --- |
| w near 1 | b is 0 and x is 2 | First entry -4 |
| b near 0 | w is 1 and x is 2 | Second entry -2 |
| Assemble in order [w, b] | Same original model | Gradient [-4, -2] |

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Guided: **prediction** 1, residual -2, loss 4, **gradient** [-8, -4]. Exit: changing both parameters measures a directional change; each partial must vary just its own coordinate while holding the others fixed.

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

### Two separate probes at one parameter vector

| PARAMETER BEING PROBED | HELD FIXED | GRADIENT ENTRY |
| --- | --- | --- |
| w near 1 | b is 0 and x is 2 | First entry -4 |
| b near 0 | w is 1 and x is 2 | Second entry -2 |
| Assemble in order [w, b] | Same original model | Gradient [-4, -2] |

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

**At w = 0, b = 1, find prediction, residual, loss and the gradient using [4 times residual, 2 times residual].**

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Gradients with a numerical check

Check each coordinate of a **gradient** while holding the other parameter fixed.

### The facts we will use

#### Supplied facts

Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/10

# Gradients with a numerical check

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Check w direction

Analytic derivative2 x residual x 2=-4.

## 2. Check b direction

Analytic derivative2 x residual=-2.

## 3. Numerical check

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

**CHANGE ONE THING**

Swap the two gradient components in the program output.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Check w direction | Analytic derivative2×residual×2=-4. |
| 2. Check b direction | Analytic derivative2×residual=-2. |
| 3. Numerical check | With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows


### Why the result follows


A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

### What this example does not establish

### Keep the stated limit

The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

**Change one thing in the pictured case**

### Change one thing

Change: Swap the two **gradient** components in the program **output**.

### Predicted result

**Prediction**: The coordinate-wise numerical check exposes the mismatch.

### Why the result follows

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

**Ask while showing the picture**

### Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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## Demonstrate and guide practice

> Demonstrate Gradients with a numerical check, reduce support deliberately and check individual reasoning before discussing answers.

| Time or stage | Teacher action |
| --- | --- |
| Session 1 start to 7 minutes | Retrieve the symmetric denominator and compute the initial prediction and residual. |
| 7 to 18 minutes | Probe w alone with b fixed; record the two losses and slope. |
| 18 to 32 minutes | Probe b alone and assemble [-4, -2]. Complete W1 and W2. |
| 32 to 45 minutes | Trace the optional numerical checker and the guided [0, 1] case. |
| Session 2 start to 8 minutes | Use W3 to distinguish a combined direction from two partial derivatives. |
| 8 to 20 minutes | Discuss the ReLU corner in W5 and a stated numerical tolerance. |
| 20 to 35 minutes | Collect W4 independently, requiring both pairs of losses and coordinate order. |
| 35 to 45 minutes | Use the exit response and fresh retry, then connect local gradient evidence to next week. |

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate w-probe and b-probe tables. Extension: at the lesson model [1, 0], test steps 0.1, 0.01 and 0.001 and report absolute disagreements from [-4, -2], without claiming universal **tolerance** guarantees.

### Reduce help deliberately

#### Completion practice

Completion task: A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

#### Reasoning to listen for

Teacher check: Teacher explanation: The interval is 0.2. (0.64 - 1.44) / 0.2 = -4, not -8.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Gradient**s with a numerical check. Proposed explanation: “The **gradient** is one scalar”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: It has one entry per parameter for a scalar loss, in a stated order.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Reasoned teacher answer

They estimate a **partial derivative**. Changing both coordinates would mix their effects and would not independently verify [-4,-2]. Changed-case check: The coordinate-wise numerical check exposes the mismatch.

The w perturbation changes y twice as strongly as the same b perturbation.

Changed-case reasoning: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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## Run the investigation and diagnose misconceptions

> Run the Gradients with a numerical check investigation and use the first unsupported step to choose a teaching response.

The **gradient** is one scalar: It has one entry per parameter for a scalar loss, in a stated order.

Use the updated weight for the bias probe: Both entries must refer to the same current parameter vector.

A symmetric value defines the derivative at any corner: Left and right slopes can differ; the ReLU example exposes this.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |

| What the response shows | Next teaching action |
| --- | --- |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Gradients with a numerical check questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Guided and exit questions

Guided: **prediction** 1, residual -2, loss 4, **gradient** [-8, -4]. Exit: changing both parameters measures a directional change; each partial must vary just its own coordinate while holding the others fixed.

### W1 Recover a gradient

#### Workbook question · W1

For the lesson model at w = 1 and b = 1, calculate **prediction**, residual, loss and **gradient**.

#### Teacher reasoning · W1

Teacher explanation: **Prediction** 3, residual 0, loss 0, **gradient** [0, 0].

### W2 Catch a denominator error

#### Workbook question · W2

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

#### Teacher reasoning · W2

Teacher explanation: The interval is 0.2. (0.64 - 1.44) / 0.2 = -4, not -8.

### W3 Separate a direction from a coordinate

#### Workbook question · W3

Starting at [w, b] = [1, 0], both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either **partial derivative**.

#### Teacher reasoning · W3

Teacher explanation: Residuals are -1.3 and -0.7; losses 1.69 and 0.49. Slope is -6. It measures the combined direction [1, 1], equal here to the sum of partials -4 and -2.

### W4 Check a new example independently

#### Workbook question · W4

Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x, 2 times residual].

### Teacher reasoning · W4

Teacher explanation: Residual -1 and loss 1. w probes give **prediction**s 0.7 and 1.3, losses 1.69 and 0.49, partial -6. b probes give **prediction**s 0.9 and 1.1, losses 1.21 and 0.81, partial -2. Formula **gradient** is [-6, -2].

## W5 Recognise a corner

### Workbook question · W5

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

### Teacher reasoning · W5

Teacher explanation: The value is 0.5. The left slope is 0 and right slope is 1, so the slopes disagree and the derivative at zero does not exist.

## Independent assessment and fresh retry

### Teacher key · Independent assessment and fresh retry

W4 requires numerical and formula **gradient**s [-6, -2] with losses shown. Fresh retry: x = 1, target 4, w = 2, b = 1 gives residual -1 and **gradient** [-2, -2]. Step 0.1 in either coordinate gives losses 1.21 and 0.81.

Require the fixed coordinate to be named.

## Assess independent transfer

### Independent question · withhold the key

Selected case: Use x = 3, target 2 and y = wx + b at w = 0, b = 1. With step 0.1, calculate

both numerical partials. Compare with [2 times residual times x, 2 times residual].

### Selected case answer

#### Teacher key · Selected case answer

Teacher explanation: Residual -1 and loss 1. w probes give **prediction**s 0.7 and 1.3, losses 1.69 and 0.49, partial -6. b probes give **prediction**s 0.9 and 1.1, losses 1.21 and 0.81, partial -2. Formula **gradient** is [-6, -2].

| Dimension | Evidence required |
|---|---|
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

#### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

# Connect the project, support and next lesson

> Connect Gradients with a numerical check to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **gradient**s with a numerical check **evidence** sheet to Image-pattern Network Investigation. Show how you can estimate one **partial derivative** while other parameters stay fixed.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate w-probe and b-probe tables. Extension: at the lesson model [1, 0], test steps 0.1, 0.01 and 0.001 and report absolute disagreements from [-4, -2], without claiming universal **tolerance** guarantees.

Week 11 applies **gradient**s to simultaneous parameter updates.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

### Plan the delayed return

Suggested return points in this level: ai-5/11, ai-5/13, ai-5/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |

| Record | Teacher evidence |
| --- | --- |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Gradients with a numerical check lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Estimate one **partial derivative** while other parameters stay fixed

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

### Supported practice answer

Reasoned practice answer: Teacher explanation: The interval is 0.2. (0.64 - 1.44) / 0.2 = -4, not -8.

Check every learner before discussion: Use the worked case for **Gradient**s with a numerical check. Proposed explanation: “The **gradient** is one scalar”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: It has one entry per parameter for a scalar loss, in a stated order.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: Swap the two **gradient** components in the program **output**. Require a **prediction** and a reason before comparing. Expected result: The coordinate-wise numerical check exposes the mismatch.

Reason: The w perturbation changes y twice as strongly as the same b perturbation.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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## Visual concept journey

> Show one Gradients with a numerical check graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Gradients with a numerical check: the case

AI ACADEMY / AI-5/10 · 01 · LOOK AT THE FACTS

### Gradients with a numerical check: the case

**The situation**

Check each coordinate of a gradient while holding the other parameter fixed.

**Supplied evidence**

Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

**By the end, I can explain gradients with a numerical check using a traced example and a stated limitation.**

Follow the labelled picture with the matching explanation. Open the vector diagram

Check each coordinate of a **gradient** while holding the other parameter fixed. Supplied facts: Model y=2w+b, target 3. At [w, b]=[1, 0], y = 2, residual y-target=-1; loss is residual².

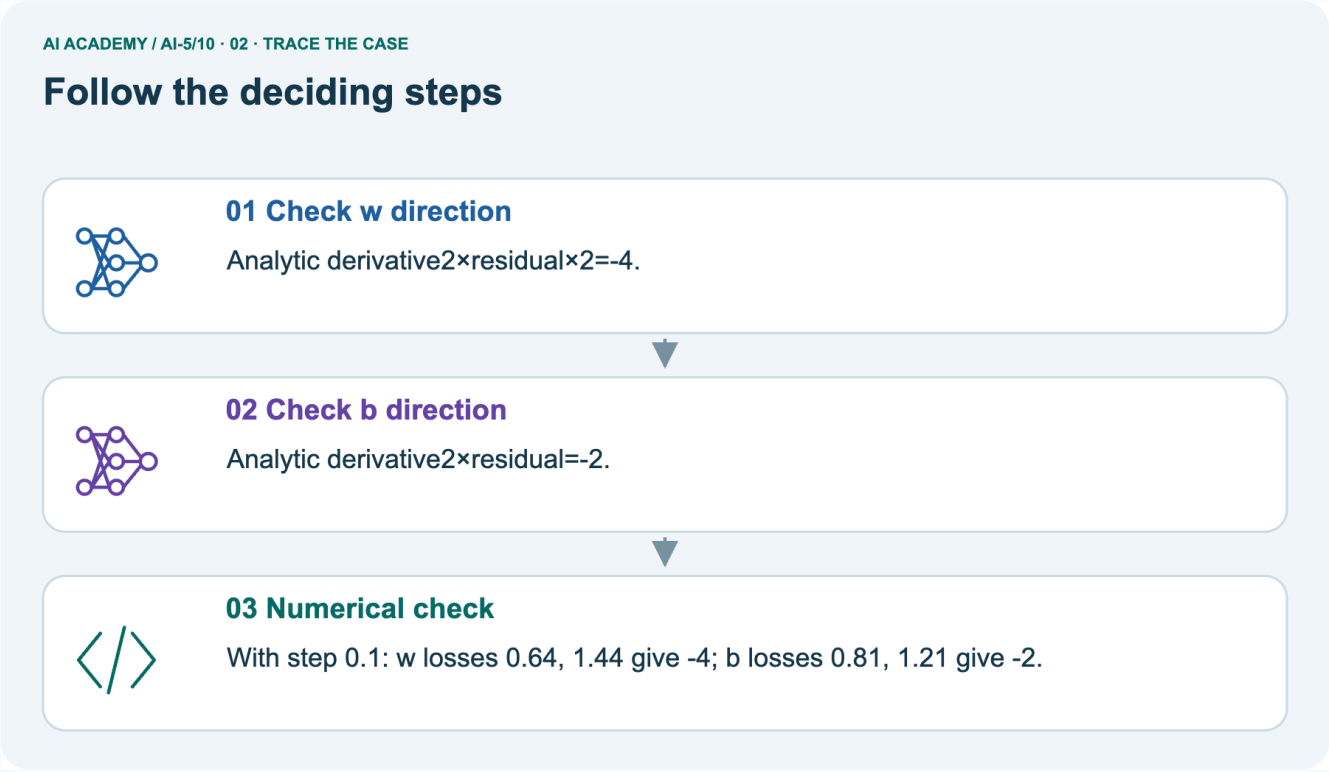
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

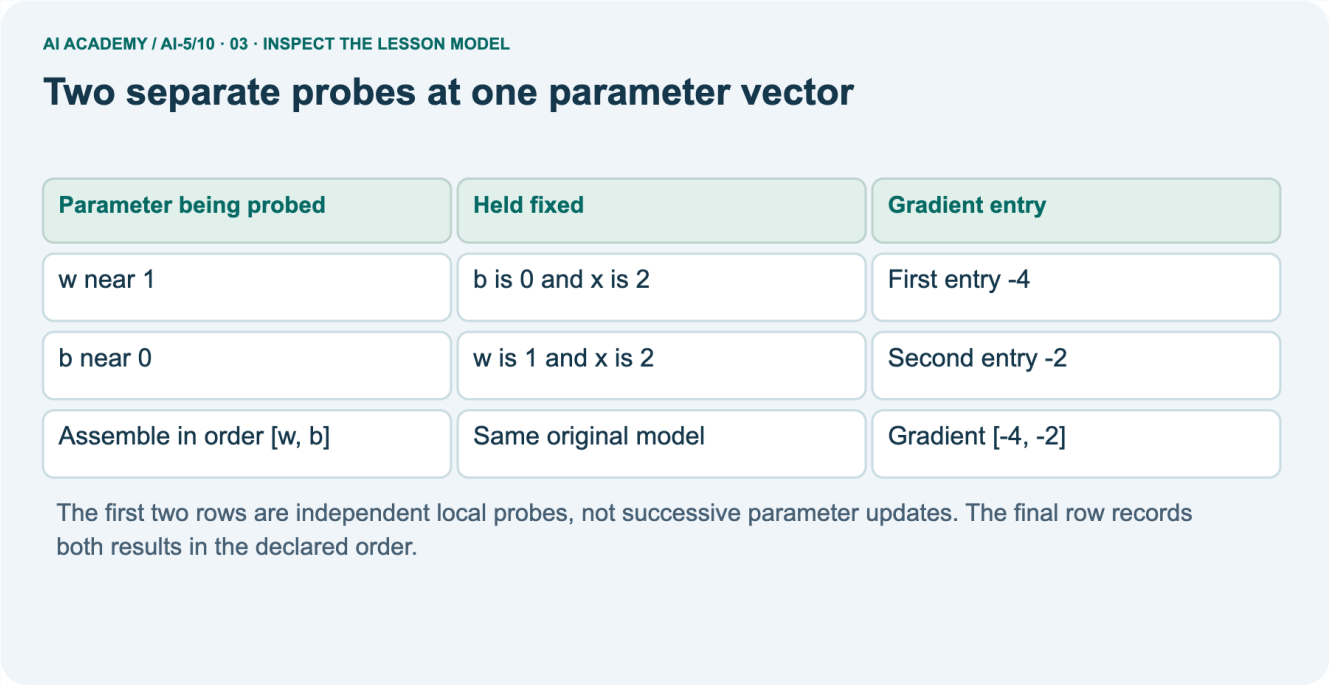
Check w direction: Analytic derivative2×residual×2=-4. → Check b direction: Analytic derivative2×residual=-2. → Numerical check: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Two separate probes at one parameter vector



AI ACADEMY / AI-5/10 · 03 · INSPECT THE LESSON MODEL

## Two separate probes at one parameter vector

| Parameter being probed | Held fixed | Gradient entry |
| --- | --- | --- |
| w near 1 | b is 0 and x is 2 | First entry -4 |
| b near 0 | w is 1 and x is 2 | Second entry -2 |
| Assemble in order [w, b] | Same original model | Gradient [-4, -2] |

The first two rows are independent local probes, not successive parameter updates. The final row records both results in the declared order.

Follow the labelled picture with the matching explanation. Open the vector diagram

Two separate probes at one parameter vector. w near 1 | b is 0 and x is 2 | First entry -4. b near 0 | w is 1 and x is 2 | Second entry -2. Assemble in order [w, b] | Same original model | **Gradient** [-4, -2]

Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 04. Change one thing. Trace what follows.

AI ACADEMY / AI-5/10 · 04 · COMPARE THE CASES

### Change one thing. Trace what follows.

**Original case**

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

**Changed condition**

Swap the two gradient components in the program output.

**Worked changed-case result**

The coordinate-wise numerical check exposes the mismatch.
Why: The w perturbation changes y twice as strongly as the same b perturbation.

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. Changed condition: Swap the two **gradient** components in the program **output**.

Expected result: The coordinate-wise numerical check exposes the mismatch. Reason: The w perturbation changes y twice as strongly as the same b perturbation.

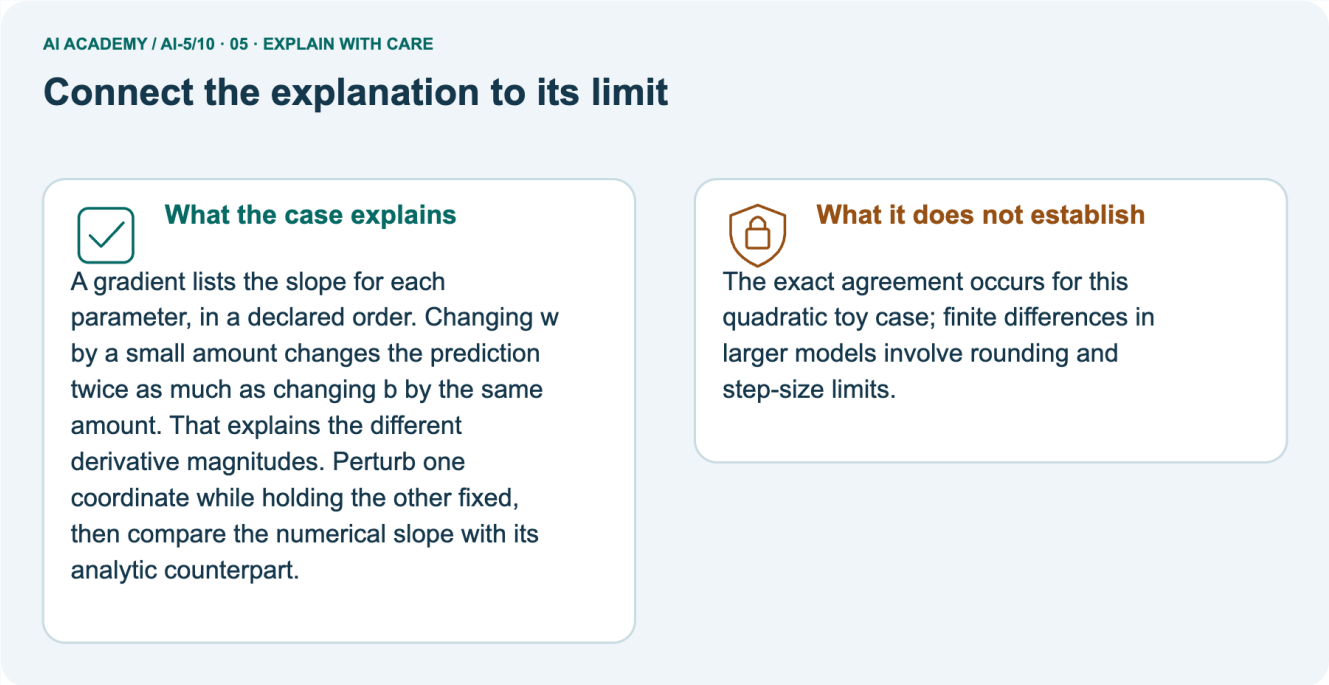
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The w perturbation changes y twice as strongly as the same b perturbation.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

A **gradient** lists the slope for each parameter, in a declared order. Changing w by a small amount changes the **prediction** twice as much as changing b by the same amount. That explains the different derivative magnitudes.

Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Limit: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

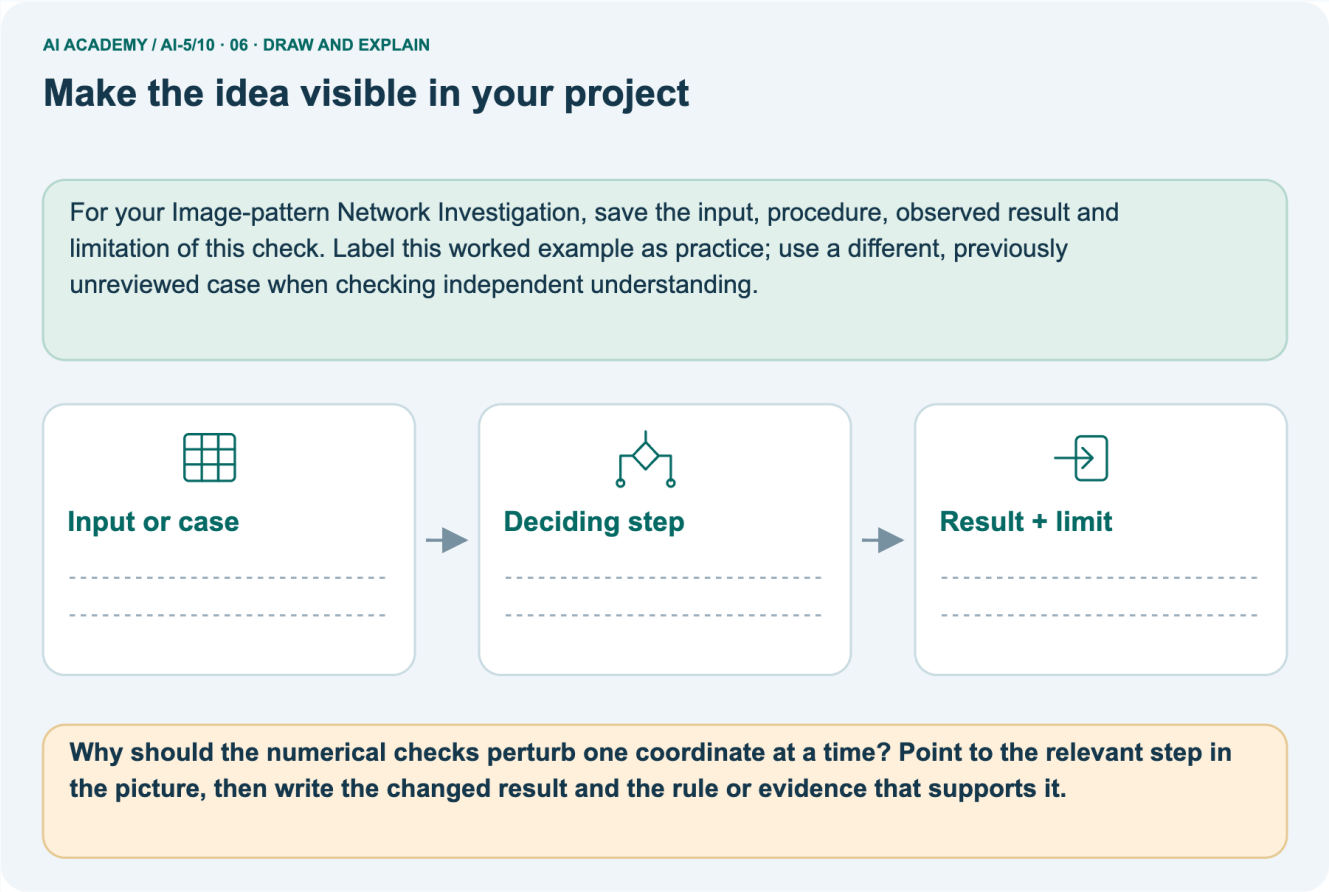
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The exact agreement occurs for this quadratic toy case; **finite difference**s in larger models involve rounding and step-size limits.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: They estimate a **partial derivative**. Changing both coordinates would mix their effects and would not independently verify [-4,-2].

Changed-case check: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.